Pranav Reddy

510-579-9141 | LinkedIn | preddy@ucsb.edu

PROJECTS & EXPERIENCE

Quantum Codesign Lab
Niu Group, UCSB Undergraduate Researcher — June 2025 — present

- Designed time-dependent Lyapunov control schedules with learnable parameters to accelerate ground state convergence in the Transverse Field Ising Model (TFIM) via feedback-based quantum algorithms (FQAs)
- Benchmarked GRU, Bayesian, and threshold decoders on continuous stabilizer readout across four hardware-realistic simulation phases; GRU achieved 96% accuracy under colored noise, measurement backaction, and drift

---

FX2Accel — January 2026 - April 2026
Language: Python

- Built multi-stage ML compiler pipeline using PyTorch FX by implementing ShapeProp, DCE, and operator fusion passes, reducing op count and eliminating intermediate memory roundtrips.
- Designed a lifetime-based memory planner for on-chip buffer reuse, mirroring memory tradeoff constraints

---

KernelForge — February 2026 - April 2026
Language: Python

- Built a Triton-inspired kernel compiler that lowers tensor ops through loop tiling (32x32) and CUDA-style codegen, mapping work across GPU thread hierarchies via block/threading indexing
- Implemented fusion and tiling optimization passes that eliminate kernel launch overhead and keep intermediate tensors in shared memory rather than global DRAM

AccelSim — February 2026 - April 2026
Language: Python

- Built a cycle accurate neural accelerator simulator modeling 5-instruction ISA with configurable latency, on-chip buffer management, and memory traffic tracking
- Implemented bottleneck detection classifying workloads as compute-bound, memory-bound, or buffer limited

---

Application of Graphs to Machine Learning
Language: C++ — January 2024–March 2024

- Implemented graph algorithms (BFS, DFS) to traverse and manage neural network structures, developed methods for node and connection management and applied graph traversal algorithms for prediction and back-propagation

EDUCATION

BS Electrical and Computer Engineering | University of California, Santa Barbara — September 2023 — Present
Pursuing electrical engineering major with an emphasis on photonics, high performance computing, computer architecture, ML/AI algorithm optimizations.

COURSE WORK

Intro Solid State Devices (ECE132)	Intro to Optimizations (ECE 133)	Probability & Stats (ECE 139)
Problem Solving in C++ (CS 24)	Signal Processing (ECE 130A)	Discrete Processing (ECE 130B)
Interface Design (ECE153B)	Digital Design (ECE152)	Waveguide Physics (ECE 144)

SKILLS & LEADERSHIP

Languages : C++, C, Python, Java, Arduino, LATEX
Tools : Unix, Git/Github, LangChain, Tavily, OpenAI, Google Cloud